APPENDIX

A1.1 ELECTRIC FIELD DUE TO CONTINUOUS CHARGE DISTRIBUTION

Consider the following charged object of irregular shape as shown in Figure A1.1. The entire charged object is divided into a large number of charge elements $\Delta q_1, \Delta q_2, \Delta q_3 ..... \Delta q_n$ and each charge element Δq is taken as a point charge.

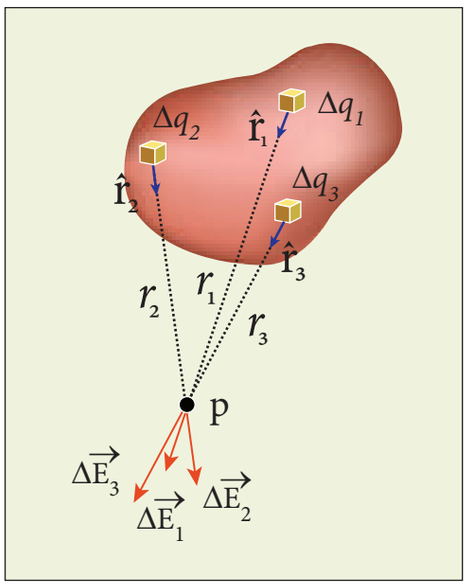


Figure A1.1 Continuous charge distributions

The electric field at a point P due to a charged object is approximately given by the sum of the fields at P due to all such charge elements.

$$\vec{E} \approx \frac{1}{4\pi\epsilon_0}\left(\frac{\Delta q_1}{r_{1P}^2}\hat{r}_{1P} + \frac{\Delta q_2}{r_{2P}^2}\hat{r}_{2P} + ... + \frac{\Delta q_n}{r_{nP}^2}\hat{r}_{nP}\right)$$
$$\approx \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{\Delta q_i}{r_{iP}^2}\hat{r}_{iP} \qquad \text{(A1.1)}$$

Here Δq_i is the ith charge element, r_{iP} is the distance of the point P from the ith charge element and $\hat{r}_{iP}$ is the unit vector from ith charge element to the point P.

However the equation (A1.1) is only an approximation. To incorporate the continuous distribution of charge, we take the limit $\Delta q \to 0 (= dq)$. In this limit, the summation in the equation (A1.1) becomes an integration and takes the following form

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\int\frac{dq}{r^2}\hat{r} \qquad \text{(A1.2)}$$

Here r is the distance of the point P from the infinitesimal charge dq and $\hat{r}$ is the unit vector from dq to point P. Even though the electric field for a continuous charge distribution is difficult to evaluate, the force experienced by some test charge q in this electric field is still given by $\vec{F} = q\vec{E}$.

(a) If the charge Q is uniformly distributed along the wire of length L, then linear charge density (charge per unit length) is $\lambda = \frac{Q}{L}$. Its unit is coulomb per meter (Cm^{-1}).

The charge present in the infinitesimal length dl is $dq = \lambda dl$. This is shown in Figure A1.2 (a).

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